

- 11 (a) Murray just said $-2 \times -4 = 8$ and possibly put it as the power in the denominator because he thought the negatives needed to be accounted for. He is wrong in at least two ways.
- (b) Raphael knows that indices are added when numbers are multiplied. $\frac{1}{a^6}$ may be Raphael's answer, which would be correct.
- (c) When numbers with indices and the same base are multiplied, keep the base the same and add the indices.
- $$a^{-2} \times a^{-4} = a^{-6} = \frac{1}{a^6}.$$

Exercise 12.7 (p 696)

- 1 (a) $x^{2/3}$ (b) $a^{5/6}$ (c) $k^{8/15}$ (d) $m^{1/4}$
 (e) $p^{1/6}$ (f) $y^{1/4}$ (g) $5p^2$ (h) $7x^3$
 (i) $2a^{1/3}b$ (j) $x^2y^{1/2}$ (k) $k^{1/2}n^4$ (l) $\frac{p^2}{q^5}$
- 2 (a) 14 (b) 23 (c) 19 (d) 8 (e) 6 (f) 8
 (g) 5 (h) 4 (i) 7.94 (j) 14.42 (k) 2.15 (l) 2.29
- 3 $4^{-1}, 6^{-\frac{1}{5}}, 2^{\frac{1}{2}}, 3^{\frac{1}{3}}$
- 4 (a) D (b) B (c) B (d) C
- 5 (a) a (b) d^3 (c) $\frac{x}{y}$ (d) $\frac{1}{m^2n^4}$
 (e) $\frac{1}{q^2}$ (f) $\frac{1}{b^4}$ (g) $\frac{m^{1/3}}{2n^{1/6}}$ (h) $\frac{y^{7/6}}{5x^{1/3}}$
- 6 (a) B (b) A
- 7 (a) 0.5 (b) 0.437 (c) 0.164 (d) 0.196
 (e) 0.369 (f) 0.343 (g) 0.398 (h) 0.359
- 8 (a) $x = \sqrt[3]{3}$ (b) $x = 4$ (c) $x = \frac{1}{2}$
- 9 (a) x^3 (b) a^3 (c) $2m^2n$
- 10 (a) $x = \pm 2$ (b) $x = 3$ (c) $x = -2$ (d) $x = 2$
- 11 (a) 7 (b) $x = 7, y = 5$ (c) 256
 (d) $d = 256, e = \frac{3}{4}$

Open-ended – Sample answers

12 $1476.106^{\frac{4}{5}}, 7^3, 1.793^{10}$

13 $\left(\frac{1}{n^4}m^{\frac{1}{5}}\right)^3, (\sqrt[4]{n} \times \sqrt[5]{m})^3, (-\sqrt[4]{n^3})(-\sqrt[5]{m^3})$

Exercise 12.8 (p. 702)

- 1 (a) $\log_2 32 = 5$ (b) $\log_2 256 = 8$
 (c) $\log_2 4096 = 12$ (d) $\log_3 27 = 3$
 (e) $\log_3 243 = 5$ (f) $\log_3 6561 = 8$
 (g) $\log_{10} 100 = 2$ (h) $\log_{10} (1\,000\,000) = 6$

(i) $\log_{10} (100\,000\,000) = 8$ (j) $\log_2 \left(\frac{1}{4}\right) = -2$

(k) $\log_2 \left(\frac{1}{8}\right) = -3$ (l) $\log_2 \left(\frac{1}{16}\right) = -4$

2 (a) $10^2 = 100$ (b) $10^3 = 1000$

(c) $10^5 = 100\,000$ (d) $2^4 = 16$

(e) $2^7 = 128$ (f) $2^9 = 512$

(g) $10^{-3} = \frac{1}{1000}$ (h) $10^{-4} = \frac{1}{10\,000}$

(i) $10^{-6} = \frac{1}{1\,000\,000}$ (j) $2^{-3} = \frac{1}{8}$

(k) $2^{-5} = \frac{1}{32}$ (l) $2^{-7} = \frac{1}{128}$

3 (a) C (b) D

4 (a) 4 (b) 8 (c) 9 (d) -2 (e) -3

(f) -6 (g) -3 (h) -5 (i) 4

5 (a) 16 (b) 64 (c) 128 (d) -4 (e) -5

(f) -8 (g) 5 (h) 7 (i) 5

6 (a) (i) 10 (ii) 5 (iii) 7

(b) $\log_a a = 1$

7 (a) (i) $x = 1$ (ii) $x = 1$ (iii) $x = 1$

(b) $\log_a 1 = 0$

8 We cannot find the square root of a negative number.

9 (a) Students to verify that both equations are true.

(b) 6

(c) $2^{12} = 4096$ is the index equivalent of the first statement and $4^y = 4096$ is the second. But, this could be written as $(2^2)^y = 4096$ or $2^{2y} = 4096$, so $2y = 12$ and $y = 6$.

Open-ended – Sample answers

10 $\log_9 (531\,441) = 6$ and $\log_{27} (531\,441) = 4$

11 Effectively Lee is saying $\log_6 (-36) = -2$, which would mean that $6^{-2} = -36$, which is clearly incorrect. Lee should have divided both sides by -1 to get $(6)^2 = 36$, which has the equivalent form $\log_6 (36) = 2$.

Exercise 12.9 (p. 708)

1 (a) $\log_2 32$ (b) $\log_2 64$ (c) $\log_2 256$

(d) $\log_{10} 30$ (e) $\log_{10} 21$ (f) $\log_{10} 36$

2 (a) $\log_2 2$ (b) $\log_2 5$ (c) $\log_2 5$

(d) $\log_{10} 14$ (e) $\log_{10} 5$ (f) $\log_{10} 6$

3 (a) $4 \log_2 5$ (b) $3 \log_2 6$ (c) $4 \log_2 7$

(d) $3 \log_{10} 7$ (e) $4 \log_{10} 8$ (f) $5 \log_{10} 11$

4 (a) 5 (b) 7 (c) 8 (d) 2 (e) 1 (f) 1

5 (a) A (b) C

6 (a) 3 (b) 11 (c) 1 (d) 2 (e) 8 (f) -9

7 (a) 2 (b) $\frac{5}{6}$ (c) 3 (d) 2 (e) $\frac{3}{2}$ (f) $\frac{4}{3}$